

Micro-Doppler Signature Analysis for Aerial Target Detection and Classification

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) have become increasingly prevalent across various applications, yet their misuse poses serious security risks. Conventional radar systems often struggle to detect and classify drones due to their small radar cross-section, low-altitude flight, and similarities to birds. This work presents a practical methodology for drone detection and classification using radar-based micro-Doppler signatures. The study combines simulated and experimental data to analyze the motion characteristics of drones and distinguish them from similar aerial objects such as birds. A MATLAB-based simulation was developed using a 3D CAD model of a two-blade rotor system to generate synthetic radar returns. Short-Time Fourier Transform analysis was applied to obtain time-frequency spectrograms and study modulation depth, blade symmetry, and the influence of rotation rate on Doppler spreading. Experimental validation was performed using a Texas Instruments IWR6843 mmWave radar module, with data collected for multiple scenarios including flying drones, static two-blade systems, birds in motion, and background clutter recordings of the measurement scene without a drone target. The collected I/Q data were preprocessed through filtering and time-frequency analysis to generate spectrograms, which were organized into labeled datasets and used to fine-tune VGG-based convolutional neural network models for classification. The proposed approach achieved 99.55% test accuracy on the test set across five classes.

Keywords: Unmanned Aerial Vehicles (UAVs), drone detection, micro-Doppler signatures, FMCW radar, mmWave sensing, deep learning, convolutional neural networks (CNN), time-frequency analysis

1. INTRODUCTION

Small unmanned aerial vehicles (UAVs) have become widely accessible for civilian and commercial use, but the same accessibility increases the likelihood of unauthorized operation. From a radar-sensing perspective, reliable UAV detection and identification remains challenging because small drones can exhibit low radar cross section, operate close to the ground, and produce returns that overlap with birds and environmental clutter when only range and bulk Doppler are considered. Recent surveys of radar-based aerial threat sensing emphasize that robust classification typically requires signatures beyond centroid velocity, particularly when targets are small and the operating environment is cluttered.¹

Micro-Doppler processing addresses this limitation by exploiting Doppler modulations induced by micro-motions superimposed on the target bulk motion. For a monostatic radar, the instantaneous Doppler shift is related to the target radial velocity $v_r(t)$ as

$$f_D(t) = \frac{2v_r(t)}{\lambda}, \quad (1)$$

where λ is the radar wavelength. Mechanical components such as rotating blades and vibrating structures introduce periodic (or quasi-periodic) variations in $v_r(t)$, which appear as structured time-frequency patterns in spectrogram representations. The physical basis, modeling framework, and time-frequency interpretation of micro-Doppler signatures have been developed extensively in the foundational work of Chen and have been consolidated in subsequent reviews and books.²⁻⁵

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A growing body of UAV-focused literature shows that propeller-driven platforms produce micro-Doppler signatures that are often more regular and harmonic than the signatures associated with flapping wings, enabling discrimination between drones and birds under appropriate sensing conditions.⁶ Early studies demonstrated feature-based classification of small UAVs and birds using micro-Doppler measurements and alignment/feature extraction pipelines, while later experimental work systematically compared drone and bird signatures across bands and target types, highlighting how rotation rate, blade geometry, and radar frequency influence the observed modulation structure.^{7–9}

A practical challenge is obtaining micro-Doppler data that are both interpretable and representative of operational sensing. Physics-based simulation is useful because it isolates the effects of rotor geometry, rotation rate, and aspect-dependent visibility, and it provides controlled test cases for validating preprocessing and time-frequency choices before moving to field measurements. Recent work has demonstrated CAD/point-cloud-based simulation pipelines that incorporate aspect-dependent visibility to generate realistic rotor returns and micro-Doppler spectrograms with experimental validation.¹⁰ In parallel, learning-based approaches increasingly treat micro-Doppler spectrograms as images and apply convolutional networks (often via transfer learning) to learn discriminative features directly from data, reducing reliance on handcrafted descriptors and improving robustness across operating conditions.^{11,12}

In this paper, we develop an end-to-end methodology for radar-based drone detection and classification that integrates point-cloud simulation of rotating rotors, experimental measurements collected using a Texas Instruments IWR6843 mmWave radar, and transfer-learning-based classification using a VGG-style convolutional network. Section 2 presents the simulation framework, measurement configuration, and spectrogram generation pipeline. Section 3 reports representative simulation and measurement results and summarizes the classification evaluation. Section 4 concludes the paper and outlines future directions.

2. METHODOLOGY

Fig. 1 summarizes the end-to-end workflow used in this study. We first generate synthetic micro-Doppler signatures using a point-cloud rotor model to understand how rotor geometry and rotation rate shape the time-frequency response. We then collect real I/Q recordings using a TI IWR6843 mmWave radar across multiple target classes (drones, birds, clutter, and noise). Both simulated and measured signals are converted into micro-Doppler spectrograms through a consistent preprocessing pipeline, and the resulting spectrograms are used to train and evaluate a transfer-learning classifier.

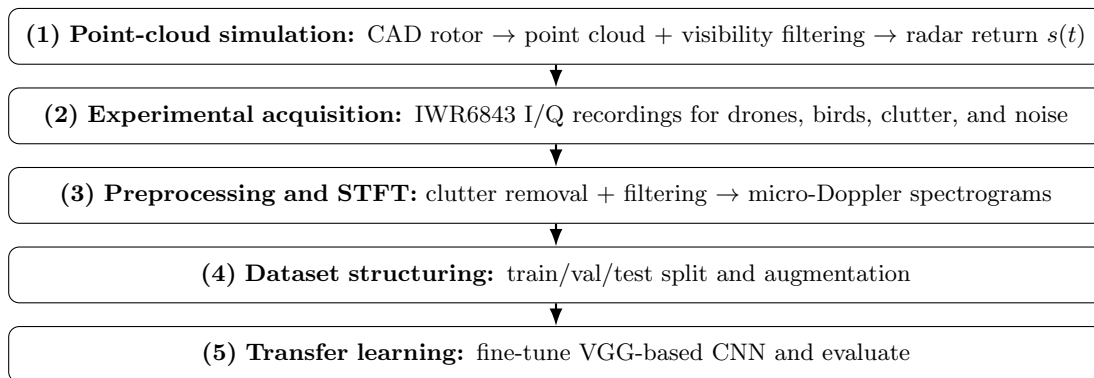


Figure 1: Overall workflow for micro-Doppler signature generation, dataset creation, and classification.

2.1 Simulation of Micro-Doppler Radar Signatures

We simulate micro-Doppler signatures by modeling a rotating rotor as a collection of discrete scatterers derived from a CAD geometry. A two-blade rotor model is first constructed (Fig. 2) and then sampled to obtain a dense point cloud representation.¹⁰ For each rotation angle, only the radar-visible points are retained using

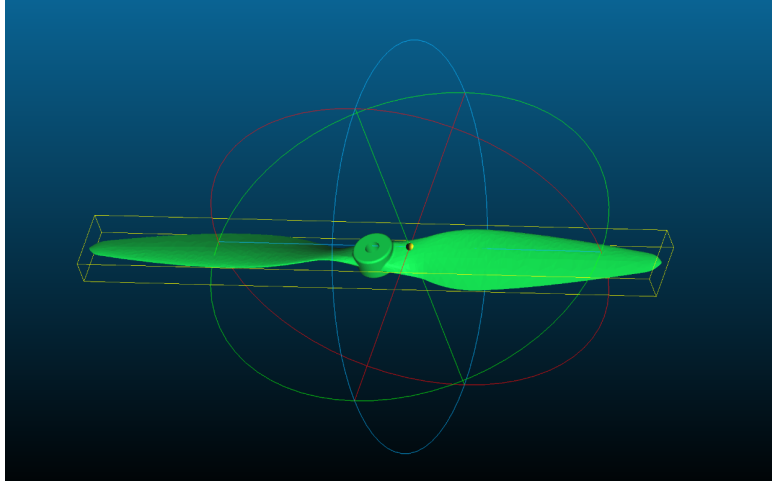


Figure 2: CAD model of two-blade rotor system.

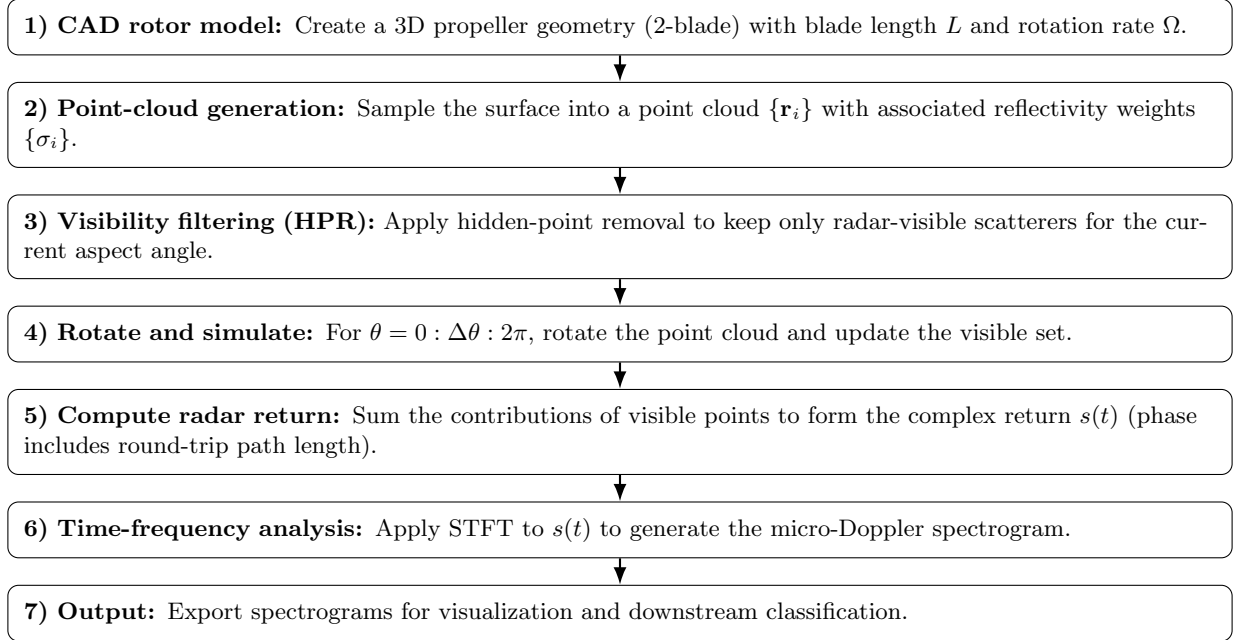


Figure 3: Point-cloud simulation methodology for generating micro-Doppler spectrograms from a rotating rotor model.

Hidden Point Removal (HPR), which approximates aspect-dependent visibility and self-shadowing. The overall point-cloud simulation workflow is summarized in Fig. 3.

Let the radar be located at $\mathbf{r}_0$ and the i th point on the rotor surface be located at $\mathbf{r}_i(\theta)$ for rotation angle θ . The corresponding range is $R_i(\theta) = \|\mathbf{r}_i(\theta) - \mathbf{r}_0\|$. The complex baseband return is modeled as a coherent sum over the visible set $\mathcal{V}(\theta)$,

$$s(\theta) = \sum_{i \in \mathcal{V}(\theta)} \sigma_i \exp\left(-j \frac{4\pi}{\lambda} R_i(\theta)\right), \quad (2)$$

where σ_i is an effective reflectivity weight and λ is the radar wavelength. As the rotor advances through small angular steps $\Delta\theta$, the sequence $s[n] \triangleq s(\theta_n)$ with $\theta_n = n\Delta\theta$ forms a time series whose phase evolution captures the micro-motion of the blades.

Time-frequency representations are obtained using the short-time Fourier transform (STFT),

$$S[m, k] = \sum_n s[n] w[n - m] e^{-j2\pi kn/N}, \quad (3)$$

where $w[\cdot]$ is a window function and N is the FFT length. The micro-Doppler spectrogram is computed as $|S[m, k]|^2$ and is used for both qualitative analysis and downstream classification.

2.2 Experimental Data Acquisition

Experimental data were acquired with a Texas Instruments IWR6843 mmWave radar Fig. 4a. Measurements were conducted across multiple target conditions, including a controlled two-blade benchtop rotor (Fig. 4b), free-flight drone runs with varying aspect angles, and background scenes containing birds and environmental clutter (Fig. 4c). The radar output consists of raw interleaved complex baseband I/Q samples saved as `.dat` files and converted into micro-Doppler spectrograms for subsequent analysis and dataset generation.

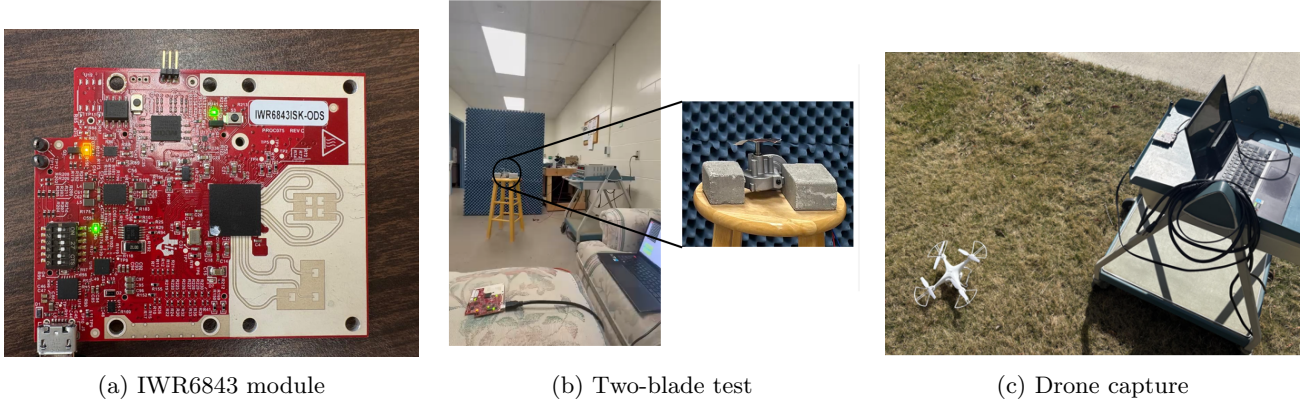


Figure 4: Experimental hardware and representative measurement setups used for micro-Doppler data collection.

2.3 Preprocessing and Spectrogram Generation

The recorded `.dat` files contain interleaved in-phase and quadrature samples that are parsed into a complex baseband sequence

$$x[n] = I[n] + jQ[n]. \quad (4)$$

For FMCW measurements, range processing is performed by taking an FFT across fast-time samples within each chirp. Let $x_\ell[n]$ denote the complex samples from chirp ℓ ; the range profile is

$$X_\ell[r] = \sum_{n=0}^{N_s-1} x_\ell[n] u[n] e^{-j2\pi rn/N_s}, \quad (5)$$

where $u[n]$ is a window and r indexes the range bin. A representative range bin is selected based on mean power over slow time,

$$r^* = \arg \max_r \frac{1}{N_c} \sum_{\ell=0}^{N_c-1} |X_\ell[r]|^2, \quad (6)$$

and the corresponding slow-time sequence is $s[\ell] \triangleq X_\ell[r^*]$. The selected slow-time sequence is then conditioned using a high-pass filter to remove residual low-frequency clutter and, when needed, a notch stage to suppress narrowband periodic interference. Finally, the micro-Doppler spectrogram is computed via the STFT in Eq. (3) and post-processed (median filtering, mild smoothing, and dynamic-range normalization) to create consistent spectrogram images for classification.

The complete preprocessing chain used to convert raw I/Q recordings into micro-Doppler spectrograms is summarized in Fig. 5.

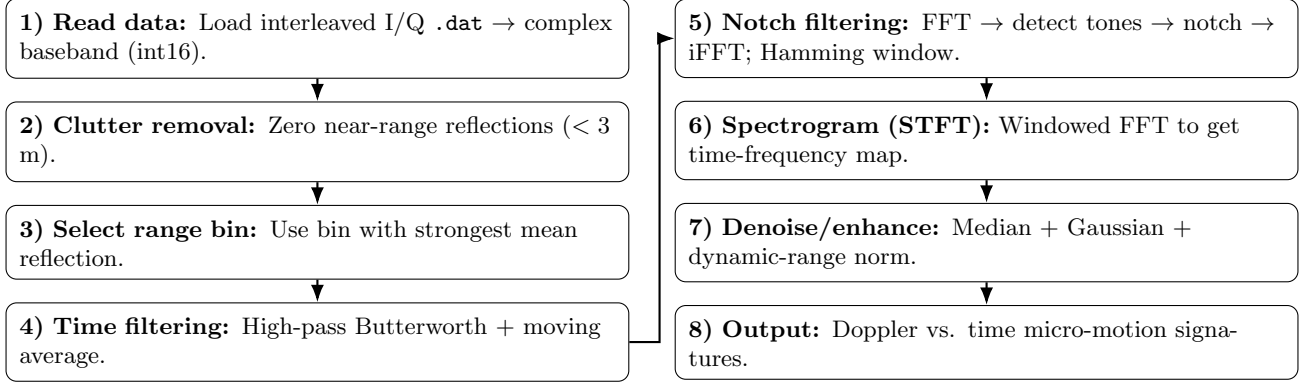


Figure 5: Processing pipeline to convert raw radar I/Q data into micro-Doppler spectrograms for classification.

2.4 Dataset Structuring

The generated spectrograms are labeled into five classes (**Drone-1**, **Drone-2**, **Birds**, **Clutter**, and **Noise**) and split into 70%/15%/15% training/validation/testing subsets. **Drone-1** is derived from the public DIAT- μ SAT dataset,¹³ whereas **Drone-2** is obtained from IWR6843 mmWave measurements collected using the procedure in Section 2.2; the **Clutter** class consists of background measurements collected with the IWR6843 in the same environments but without a drone present, capturing ground returns and static scene reflections. All classes are converted into spectrograms using the same preprocessing and normalization pipeline, with lightweight augmentation (random horizontal flipping) applied during training for consistency. To avoid information leakage, the train/validation/test split is performed at the recording level, i.e., all spectrogram frames derived from a given raw capture are assigned to a single subset. Table 1 summarizes the class distribution.

Table 1: Number of images per class in each split

Class	Train	Val	Test
Birds	982	210	211
Clutter	420	90	90
Drone-1	2263	485	486
Drone-2	84	18	18
Noise	420	90	90
Total	4169	893	895

2.5 Model Architecture and Training

We use a transfer-learning approach based on the pretrained VGG16 network, treating the convolutional blocks as a fixed feature extractor and retraining only the classifier layers for the five-class spectrogram task. Given an input spectrogram image $\mathbf{x}$, the backbone produces a feature vector $\mathbf{h} \in \mathbb{R}^{25088}$, which is mapped to class scores through a compact multilayer perceptron (MLP) with dropout regularization. The network outputs log-probabilities via a log-softmax layer, and the model is trained by minimizing the negative log-likelihood

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{n=1}^N \log p_{\theta}(y_n | \mathbf{x}_n), \quad (7)$$

where $y_n \in \{1, \dots, K\}$ with $K = 5$. Optimization is performed using Adam on the classifier parameters, with a step-based learning-rate schedule, as summarized in Algorithm 1.

Algorithm 1 Transfer-learning pipeline for micro-Doppler classification (Modified VGG16)

Require: Labeled spectrogram dataset $\mathcal{D}$ (K classes), split ratio (0.70, 0.15, 0.15), epochs E , batch size B

Ensure: Best model weights θ^* and test metrics

- 1: Split $\mathcal{D} \rightarrow \{\mathcal{D}_{tr}, \mathcal{D}_{val}, \mathcal{D}_{te}\}$; resize to 224×224 ; normalize (ImageNet); augment train with random horizontal flip
 - 2: Load pretrained VGG16; freeze feature extractor; replace classifier with $\text{FC} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{FC} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{FC} (\rightarrow K) + \text{LogSoftmax}$
 - 3: Set loss: NLLLoss; optimizer: Adam (classifier params); scheduler: StepLR
 - 4: $a^* \leftarrow 0$
 - 5: **for** $e = 1$ to E **do**
 - 6: Train on $\mathcal{D}_{tr}$ (update θ) and evaluate on $\mathcal{D}_{val}$ (no updates)
 - 7: Update scheduler; if $a_{val} > a^*$ then save $\theta^* \leftarrow \theta$, $a^* \leftarrow a_{val}$
 - 8: **end for**
 - 9: Evaluate θ^* on $\mathcal{D}_{te}$ and report accuracy, per-class metrics, and confusion matrix
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3. RESULTS AND DISCUSSION

This section presents the results obtained from simulations, experimental data collection, and machine learning classification. The analysis is focused on identifying drones using micro-Doppler signatures captured from both simulated and real-world measurements. Each subsection provides detailed observations with visual support from the spectrograms and model evaluation plots.

3.1 Analysis of Simulation Results

The micro-Doppler behavior of rotating blades was analyzed using the point-cloud simulation described in Section 2.1 and the micro-Doppler spectrogram generated by the point-cloud simulation is shown in Fig. 6. For the two-blade rotor model, the spectrogram exhibits a strongly periodic modulation with approximately symmetric positive and negative Doppler excursions about the centroid, as expected for a rigid rotating scatterer. The simulation provides a controlled baseline for interpreting the measured signatures and verifying that the preprocessing and STFT settings preserve rotor micro-motion structure.

The frequency of the Doppler shift f_D is directly related to the blade tip speed v as

$$f_D = \frac{2v}{\lambda}. \quad (8)$$

The blade tip speed is given by

$$v = \omega L, \quad (9)$$

where λ is the radar wavelength, ω is the angular velocity, and L is the blade length. The time interval T_c between consecutive peaks in the spectrogram corresponds to one rotation cycle, so the rotation rate is

$$f_r = \frac{1}{T_c}, \quad (10)$$

and the angular velocity follows as

$$\omega = 2\pi f_r = \frac{2\pi}{T_c}. \quad (11)$$

3.2 Experimental Observations from 2-Motor Setup

Fig. 7 shows a representative Micro-Doppler spectrogram measured from the controlled two-blade rotor setup (see Fig. 4b). As expected for a rigid rotating reflector, the response is strongly periodic and approximately symmetric about the Doppler centroid, with alternating positive and negative excursions as the blades move toward and away from the radar. The Doppler span and the temporal repetition rate provide interpretable proxies for blade-tip speed and rotation rate, whereas the faint notches/sidebands are consistent with narrowband interference and the windowing/conditioning stages in the preprocessing chain. This controlled dataset is therefore used to verify the STFT and normalization settings before evaluating free-flight measurements.

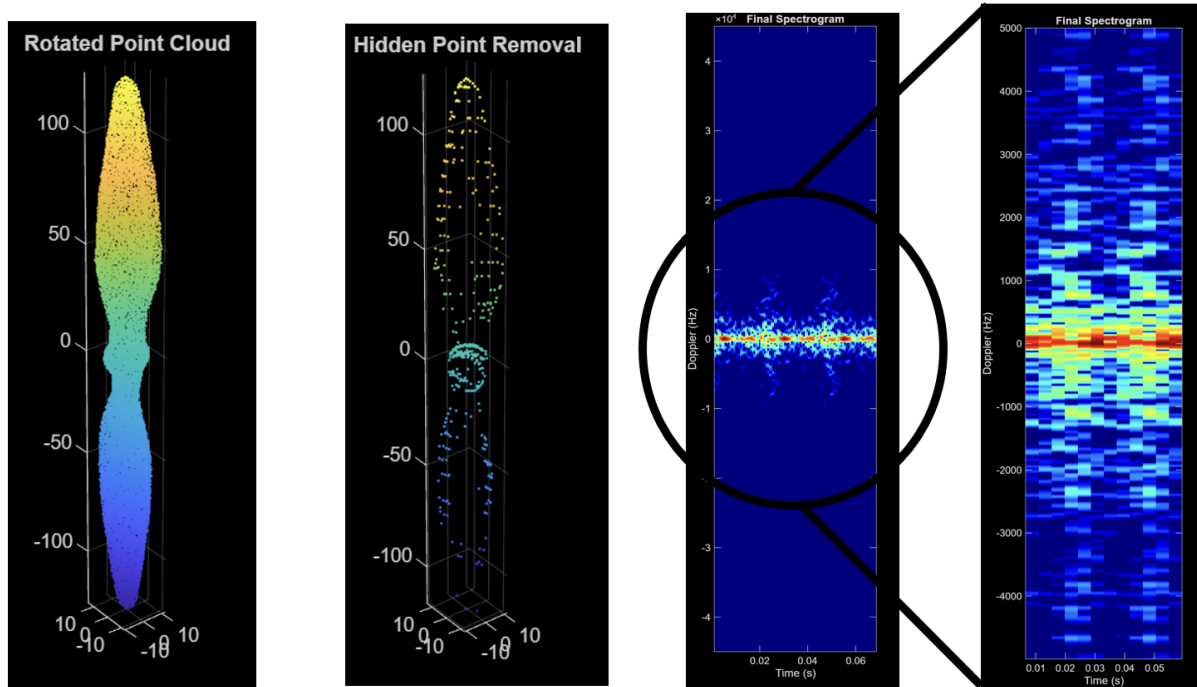


Figure 6: Point-cloud based Micro-Doppler simulation output.

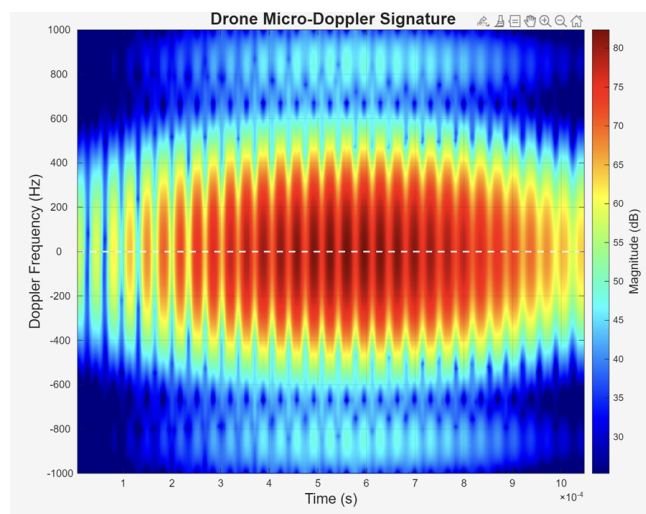


Figure 7: Micro-Doppler signatures from the 2-Blade system.

3.3 Experimental Observations from Drone Measurements

Data collected from free-flight drone measurements using the TI IWR6843 mmWave radar exhibit a persistent periodic micro-Doppler component associated with propeller rotation (Fig. 8). Compared to the controlled benchtop rotor case, the flight signatures are less symmetric about the Doppler centroid due to time-varying aspect angle (yaw/roll/pitch), platform translation, and rotor-to-rotor speed differences during maneuvering. Even with these variations, the signature remains structured and repeatable, which makes it useful for drone classification.

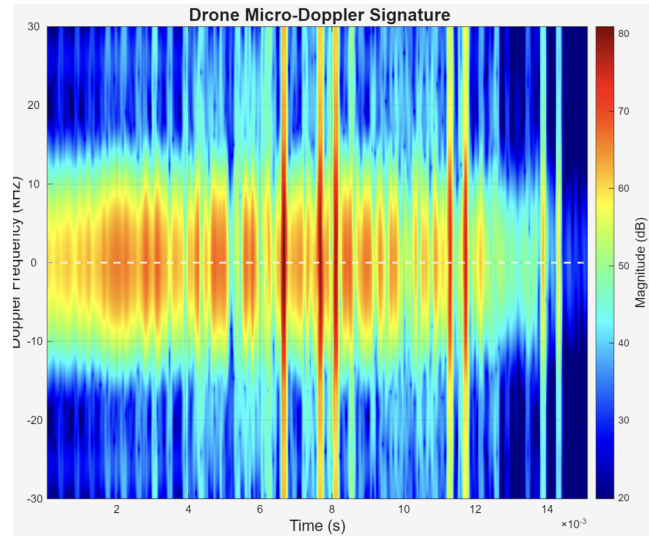


Figure 8: Representative Micro-Doppler spectrogram from a drone flight measurement.

3.4 Comparison with Bird Micro-Motions

Bird measurements were collected for comparison using the same IWR6843 radar setup as drone experiments. Data were recorded at outdoor campus locations (open fields and parks near trees) by pointing the radar toward areas with bird activity and recording until a bird pass was captured. Species were not identified; the recordings include different flight directions, including crossing and approach/recede motion, as well as birds moving between the sky, ground, and nearby trees. Portions of the recordings affected by motion of the setup were removed during preprocessing.

As shown in Fig. 9, the resulting micro-Doppler signatures are less regular and more scattered than those of drones. This behavior is consistent with flapping-wing motion, which is non-periodic compared to the steady mechanical rotation of drone propellers. Bird spectrograms also vary with flight behavior and aspect angle, which contributes to the challenge of separating birds from drones using only range or bulk Doppler features.

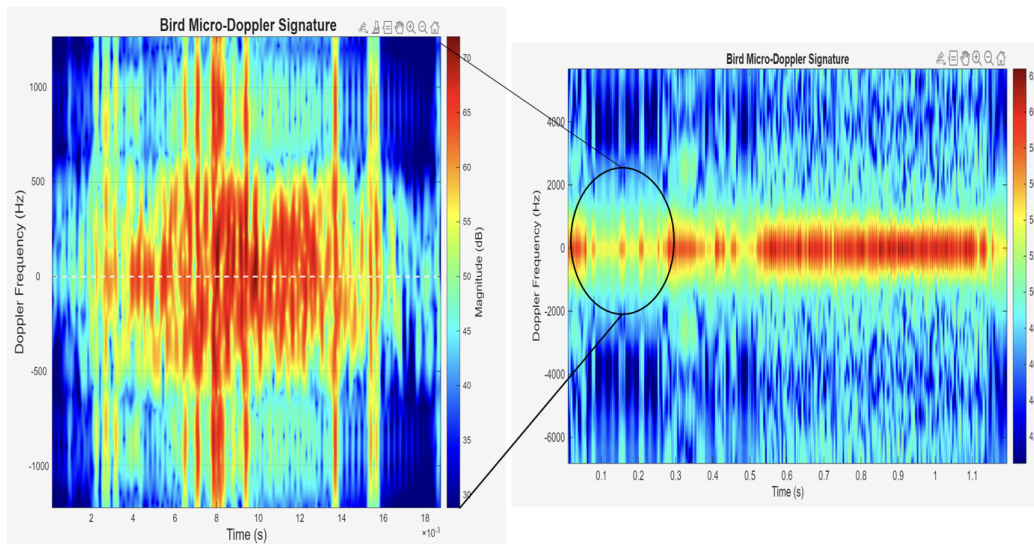


Figure 9: Representative Micro-Doppler spectrogram measured from birds in motion.

3.5 Machine Learning Model Performance

The final classification stage treats each micro-Doppler spectrogram as an image input to a convolutional neural network. All spectrograms are resized to 224×224 and normalized prior to inference. A VGG16 backbone pretrained on ImageNet is used as a feature extractor, and a lightweight classifier head with dropout produces $K = 5$ class log-probabilities. During inference, the predicted class is obtained from the maximum a posteriori estimate $\hat{y} = \arg \max_k p(y = k | \mathbf{x})$.

The model was trained on a dataset split in the ratio of 70:15:15 for training, validation, and testing respectively. Data augmentation techniques such as random horizontal flipping were employed.

Overall performance is summarized in Tables 2 and 3, demonstrating strong classification performance across the five target classes. Figure 10 shows that most errors occur between **Birds** and **Clutter**, which is expected because both can produce broadband, low-SNR time-frequency textures without strong periodic rotor harmonics. The two drone classes are classified with near-perfect accuracy, indicating that the model captures rotor-driven periodic structure in the spectrogram representation. Because **Drone-1** and **Drone-2** originate from different measurement sources, part of the separability may reflect domain differences; future work will evaluate cross-source generalization by training on one source and testing on the other.

Table 2: Overall model performance

Split	Accuracy (%)	Loss
Training	99.81	0.0045
Validation	99.33	0.0355
Test	99.55	—

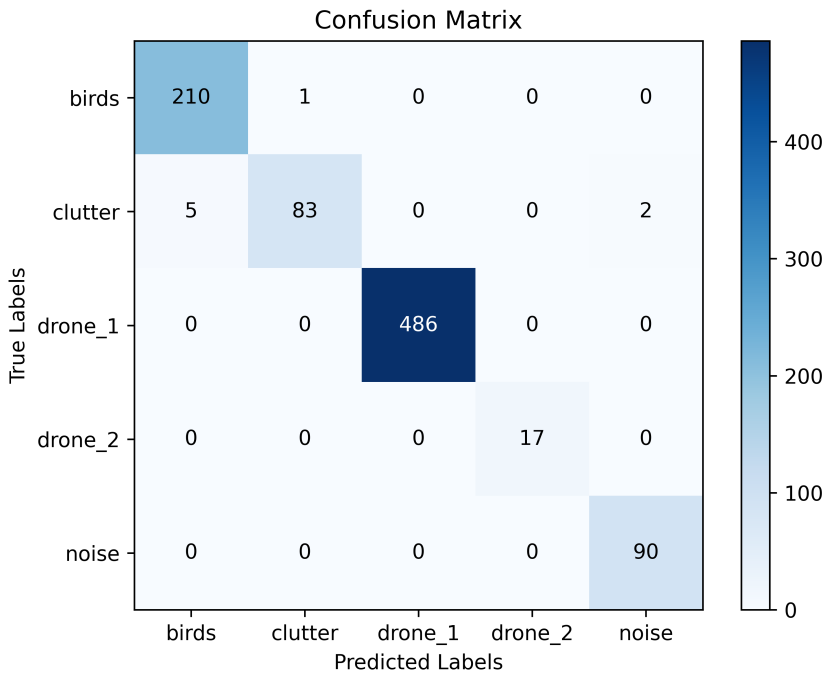


Figure 10: Confusion matrix (counts) on the test set. Misclassifications are primarily between **Birds** and **Clutter**, while both drone classes and **Noise** are classified with near-perfect accuracy.

Table 3: Per-class performance on the test set

Class	Precision	Recall	F1-score	Support
Birds	0.99	1.00	0.99	211
Clutter	0.99	0.97	0.98	90
Drone-1	1.00	1.00	1.00	486
Drone-2	1.00	1.00	1.00	18
Noise	1.00	1.00	1.00	90
Macro avg	0.99	0.99	0.99	895
Weighted avg	1.00	1.00	1.00	895

4. CONCLUSION

This work presented a practical end-to-end methodology for drone detection and classification using radar micro-Doppler signatures. A point-cloud simulation framework was used to generate synthetic returns from rotating rotors and to connect rotor geometry and rotation rate to observable time-frequency structure. Experimental I/Q measurements collected with an IWR6843 mmWave radar were processed through a consistent clutter suppression, interference mitigation, and STFT pipeline to form labeled micro-Doppler spectrogram datasets. Using transfer learning with a VGG16 backbone, the proposed classifier achieved 99.55% test accuracy across five classes (drones, birds, clutter, and noise), with near-perfect per-class precision and recall.

Future work will focus on expanding the diversity of rotor types, flight conditions, and background scenes to better quantify generalization under class imbalance and domain shift, and on implementing the processing and classification pipeline in real time for operational sensing.

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